

# Extrema and the Mean Value Theorem



## Critical points and extrema

- 1 Find the critical points of  $f(x) = 2x^3 - 3x^2 - 12x + 5$ .
  - 2 Find the absolute maximum and minimum of  $f(x) = x^3 - 3x$  on  $[-2, 3]$ , checking critical points and endpoints.
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## The Extreme Value Theorem

- 3 State the Extreme Value Theorem and explain why it guarantees  $f(x) = x^2 - 4x$  has an absolute max and min on  $[0, 5]$ .
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## Rolle's Theorem

- 4 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 4x + 3$  on  $[1, 3]$ , and find the value(s) of  $c$  guaranteed by the theorem.
  - 5 Explain why Rolle's Theorem does NOT apply to  $f(x) = |x - 2|$  on  $[0, 4]$ , even though  $f(0) = f(4) = 2$ .
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## The Mean Value Theorem

- 6 For  $f(x) = x^3$  on  $[0, 2]$ , find the value(s) of  $c$  guaranteed by the Mean Value Theorem.
  - 7 A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.
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## Using the MVT to bound function values

- 8 Given  $f(2) = 5$  and  $-3 \leq f'(x) \leq 4$  for all  $x$ , use the MVT to find the range of possible values for  $f(6)$ .
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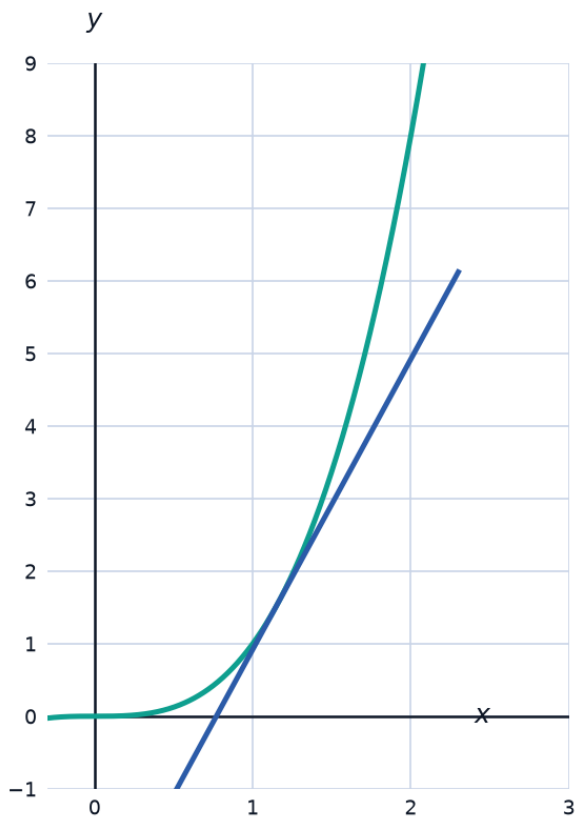
## Consequences of the MVT: monotonicity

- 9 Explain, using the MVT, why  $f'(x) > 0$  on an interval implies  $f$  is strictly increasing on that interval.

- 10 Explain why two functions  $f$  and  $g$  with  $f'(x) = g'(x)$  for all  $x$  must differ by a constant, using the MVT.
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### Visualizing the Mean Value Theorem

- 11 The graph shows  $f(x) = x^3$  on  $[0, 2]$  together with the secant line through the endpoints and the tangent line at  $x = \frac{2}{\sqrt{3}}$  (found earlier).



### Absolute extrema on a closed interval — a capstone problem

- 12 Find the absolute maximum and minimum of  $f(x) = x^4 - 4x^2 + 3$  on  $[-2, 2]$ , checking all critical points and endpoints.
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### Rolle's Theorem with trigonometric functions

- 13 Verify that Rolle's Theorem applies to  $f(x) = \sin x$  on  $[0, \pi]$ , and find the value of  $c$  guaranteed by the theorem.
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## An applied Mean Value Theorem problem

- 14 A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.
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## Multiple choice — easy 15

- 15 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 4x$  on  $[-6, 2]$ , and find the value of  $C$  guaranteed by the theorem.
- a.  $-2$
  - b.  $-6$
  - c.  $-1$
  - d.  $2$
- 16 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 6x$  on  $[1, 5]$ , and find the value of  $C$  guaranteed by the theorem.
- a.  $3$
  - b.  $4$
  - c.  $1$
  - d.  $5$
- 17 Verify that Rolle's Theorem applies to  $f(x) = x^2$  on  $[-2, 2]$ , and find the value of  $C$  guaranteed by the theorem.
- a.  $1$
  - b.  $0$
  - c.  $2$
  - d.  $-2$

- 18 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 10x$  on  $[-11, 1]$ , and find the value of  $c$  guaranteed by the theorem.
- a.  $-5$
  - b.  $1$
  - c.  $-4$
  - d.  $-11$
- 19 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 10x$  on  $[0, 10]$ , and find the value of  $c$  guaranteed by the theorem.
- a.  $5$
  - b.  $10$
  - c.  $6$
  - d.  $0$
- 20 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 8x$  on  $[-1, 9]$ , and find the value of  $c$  guaranteed by the theorem.
- a.  $5$
  - b.  $9$
  - c.  $4$
  - d.  $-1$
- 21 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 4x$  on  $[-5, 1]$ , and find the value of  $c$  guaranteed by the theorem.
- a.  $-1$
  - b.  $-2$
  - c.  $-5$
  - d.  $1$

- 22 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 10x$  on  $[2, 8]$ , and find the value of  $c$  guaranteed by the theorem.
- a. 5
  - b. 6
  - c. 2
  - d. 8
- 23 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 12x$  on  $[-9, -3]$ , and find the value of  $c$  guaranteed by the theorem.
- a. -5
  - b. -6
  - c. -3
  - d. -9
- 24 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 2x$  on  $[-2, 0]$ , and find the value of  $c$  guaranteed by the theorem.
- a. -2
  - b. 2
  - c. -1
  - d. 0
- 25 Verify that Rolle's Theorem applies to  $f(x) = x^2 + 10x$  on  $[-6, -4]$ , and find the value of  $c$  guaranteed by the theorem.
- a. -3
  - b. -6
  - c. -4
  - d. -5

- 26 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 12x$  on  $[0, 12]$ , and find the value of  $c$  guaranteed by the theorem.
- a. 0
  - b. 12
  - c. 7
  - d. 6
- 27 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 6x$  on  $[2, 4]$ , and find the value of  $c$  guaranteed by the theorem.
- a. 4
  - b. 5
  - c. 2
  - d. 3
- 28 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 8x$  on  $[3, 5]$ , and find the value of  $c$  guaranteed by the theorem.
- a. 3
  - b. 7
  - c. 4
  - d. 5
- 29 Verify that Rolle's Theorem applies to  $f(x) = x^2 - 12x$  on  $[1, 11]$ , and find the value of  $c$  guaranteed by the theorem.
- a. 6
  - b. 1
  - c. 7
  - d. 11

### Multiple choice — medium 15

- 30 Given  $f(-3) = -2$  and  $-4 \leq f'(x) \leq 6$  for all  $x$ , use the MVT to find the largest possible value of  $f(3)$ .
- a.  $-26$
  - b.  $40$
  - c.  $34$
  - d.  $28$
- 31 Given  $f(1) = -5$  and  $-4 \leq f'(x) \leq 4$  for all  $x$ , use the MVT to find the largest possible value of  $f(7)$ .
- a.  $-29$
  - b.  $25$
  - c.  $15$
  - d.  $19$
- 32 Given  $f(-1) = -2$  and  $-6 \leq f'(x) \leq 6$  for all  $x$ , use the MVT to find the largest possible value of  $f(3)$ .
- a.  $26$
  - b.  $22$
  - c.  $-26$
  - d.  $16$
- 33 Given  $f(-3) = 6$  and  $-4 \leq f'(x) \leq 5$  for all  $x$ , use the MVT to find the largest possible value of  $f(5)$ .
- a.  $41$
  - b.  $46$
  - c.  $-26$
  - d.  $54$

- 34 Given  $f(3) = -8$  and  $-1 \leq f'(x) \leq 3$  for all  $x$ , use the MVT to find the largest possible value of  $f(11)$ .
- a. 13
  - b. 16
  - c. -16
  - d. 24
- 35 Given  $f(2) = 1$  and  $-5 \leq f'(x) \leq 4$  for all  $x$ , use the MVT to find the largest possible value of  $f(9)$ .
- a. -34
  - b. 36
  - c. 25
  - d. 29
- 36 Given  $f(-4) = -4$  and  $-6 \leq f'(x) \leq 2$  for all  $x$ , use the MVT to find the largest possible value of  $f(4)$ .
- a. -52
  - b. 10
  - c. 20
  - d. 12
- 37 Given  $f(1) = -7$  and  $-3 \leq f'(x) \leq 6$  for all  $x$ , use the MVT to find the largest possible value of  $f(9)$ .
- a. 41
  - b. 35
  - c. 49
  - d. -31

- 38 Given  $f(-2) = 7$  and  $-6 \leq f'(x) \leq 4$  for all  $x$ , use the MVT to find the largest possible value of  $f(-1)$ .
- a. 11
  - b. 12
  - c. 1
  - d. 8
- 39 Given  $f(4) = 8$  and  $-3 \leq f'(x) \leq 4$  for all  $x$ , use the MVT to find the largest possible value of  $f(6)$ .
- a. 12
  - b. 18
  - c. 16
  - d. 2
- 40 Given  $f(1) = -5$  and  $-2 \leq f'(x) \leq 2$  for all  $x$ , use the MVT to find the largest possible value of  $f(2)$ .
- a. 0
  - b.  $-7$
  - c.  $-3$
  - d.  $-2$
- 41 Given  $f(-3) = 3$  and  $-3 \leq f'(x) \leq 5$  for all  $x$ , use the MVT to find the largest possible value of  $f(-1)$ .
- a. 8
  - b. 15
  - c. 13
  - d.  $-3$

- 42 Given  $f(-4) = -3$  and  $-6 \leq f'(x) \leq 2$  for all  $x$ , use the MVT to find the largest possible value of  $f(2)$ .
- a. 9
  - b. 7
  - c. -39
  - d. 15
- 43 Given  $f(1) = 1$  and  $-2 \leq f'(x) \leq 5$  for all  $x$ , use the MVT to find the largest possible value of  $f(4)$ .
- a. -5
  - b. 19
  - c. 16
  - d. 11
- 44 Given  $f(3) = 5$  and  $-2 \leq f'(x) \leq 3$  for all  $x$ , use the MVT to find the largest possible value of  $f(6)$ .
- a. 14
  - b. -1
  - c. 11
  - d. 17

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### Multiple choice — hard 15

- 45 Find the larger critical point of  $f(x) = x^3 - 9x^2 + 24x$ .
- a. 4
  - b. 2
  - c. 6
  - d. 8

46 Find the larger critical point of  $f(x) = x^3 + 9x^2 + 24x$ .

a.  $-4$

b.  $-2$

c.  $8$

d.  $-6$

47 Find the larger critical point of  $f(x) = x^3 + 6x^2 - 36x$ .

a.  $-4$

b.  $2$

c.  $-6$

d.  $-12$

48 Find the larger critical point of  $f(x) = x^3 - 108x$ .

a.  $-36$

b.  $6$

c.  $0$

d.  $-6$

49 Find the larger critical point of  $f(x) = x^3 + 9x^2$ .

a.  $-2$

b.  $3$

c.  $-6$

d.  $0$

- 50 Given  $-3 \leq f'(x) \leq 2$  for all  $x$ , use the MVT to find the width of the range of possible values for  $f(9) - f(2)$ .
- a. 31
  - b. 5
  - c. 42
  - d. 35
- 51 Given  $-5 \leq f'(x) \leq 6$  for all  $x$ , use the MVT to find the width of the range of possible values for  $f(1) - f(0)$ .
- a. 11
  - b. 1
  - c. 14
  - d. 12
- 52 Given  $-2 \leq f'(x) \leq 5$  for all  $x$ , use the MVT to find the width of the range of possible values for  $f(8) - f(3)$ .
- a. 35
  - b. 7
  - c. 15
  - d. 40
- 53 Given  $-2 \leq f'(x) \leq 1$  for all  $x$ , use the MVT to find the width of the range of possible values for  $f(2) - f(-2)$ .
- a. 3
  - b. 12
  - c. 15
  - d. 16

- 54 Given  $-3 \leq f'(x) \leq 2$  for all  $x$ , use the MVT to find the width of the range of possible values for  $f(5) - f(-2)$ .
- a. 32
  - b. 42
  - c. 35
  - d. 5
- 55 Does Rolle's Theorem guarantee a  $c \in (-4, 6)$  with  $f'(c) = 0$  for  $f(x) = (x - 1)^2$  on  $[-4, 6]$ ?
- a. Yes, but only because  $f$  is increasing on the whole interval.
  - b. Yes —  $f(-4) = f(6) = 25$ , and  $f$  is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here  $c = 1$ .
  - c. No — although  $f(-4) = f(6)$ ,  $f$  is not differentiable on  $(-4, 6)$ , so Rolle's Theorem does not apply.
  - d. No — Rolle's Theorem never applies to functions of the form  $(x - 1)^2$ .
- 56 Does Rolle's Theorem guarantee a  $c \in (-7, 5)$  with  $f'(c) = 0$  for  $f(x) = |x + 1|$  on  $[-7, 5]$ ?
- a. Yes —  $f(-7) = f(5) = 6$ , and  $f$  is continuous and differentiable everywhere, so Rolle's Theorem applies; here  $c = -1$ .
  - b. No — Rolle's Theorem never applies to absolute value functions, regardless of the interval.
  - c. No — although  $f(-7) = f(5) = 6$ ,  $f$  has a corner at  $x = -1 \in (-7, 5)$  and is not differentiable there, so one hypothesis of Rolle's Theorem fails.
  - d. Yes, since  $f(-7) = f(5)$  is the only hypothesis Rolle's Theorem requires.
- 57 Does Rolle's Theorem guarantee a  $c \in (-7, 5)$  with  $f'(c) = 0$  for  $f(x) = (x + 1)^2$  on  $[-7, 5]$ ?
- a. Yes, but only because  $f$  is increasing on the whole interval.
  - b. No — Rolle's Theorem never applies to functions of the form  $(x + 1)^2$ .
  - c. Yes —  $f(-7) = f(5) = 36$ , and  $f$  is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here  $c = -1$ .
  - d. No — although  $f(-7) = f(5)$ ,  $f$  is not differentiable on  $(-7, 5)$ , so Rolle's Theorem does not apply.

- 58** Does Rolle's Theorem guarantee a  $c \in (-5, 5)$  with  $f'(c) = 0$  for  $f(x) = (x)^2$  on  $[-5, 5]$ ?
- a.** Yes —  $f(-5) = f(5) = 25$ , and  $f$  is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here  $c = 0$ .
  - b.** Yes, but only because  $f$  is increasing on the whole interval.
  - c.** No — although  $f(-5) = f(5)$ ,  $f$  is not differentiable on  $(-5, 5)$ , so Rolle's Theorem does not apply.
  - d.** No — Rolle's Theorem never applies to functions of the form  $(x)^2$ .
- 59** Does Rolle's Theorem guarantee a  $c \in (2, 6)$  with  $f'(c) = 0$  for  $f(x) = |x - 4|$  on  $[2, 6]$ ?
- a.** No — Rolle's Theorem never applies to absolute value functions, regardless of the interval.
  - b.** Yes, since  $f(2) = f(6)$  is the only hypothesis Rolle's Theorem requires.
  - c.** No — although  $f(2) = f(6) = 2$ ,  $f$  has a corner at  $x = 4 \in (2, 6)$  and is not differentiable there, so one hypothesis of Rolle's Theorem fails.
  - d.** Yes —  $f(2) = f(6) = 2$ , and  $f$  is continuous and differentiable everywhere, so Rolle's Theorem applies; here  $c = 4$ .